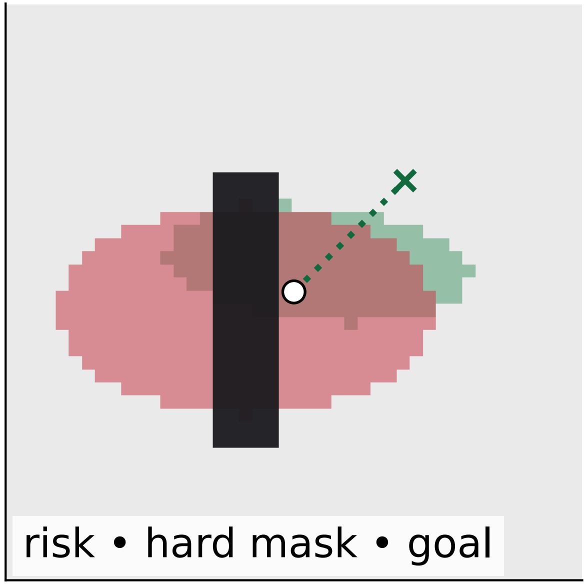
Material-aware port-Hamiltonian navigation

1 Local BEV percept



2 Feasibility witness

sample K short maneuvers

traversable + clearance

risk improvement $\Rightarrow g$

3 Additive energies

$H = H_{\mathrm{kin}} + H_{\mathrm{geom}}$

$+ \lambda_s \tilde{r} + \lambda_h b(\phi)$

heads predict λ_s, λ_h

4 Force channels

$f_{\mathrm{mat}} = -\nabla \tilde{r}$

$f_{\mathrm{haz}} = -b'(\phi) \nabla \phi$

$F = f_{\mathrm{geom}} + g \lambda_s f_{\mathrm{mat}} + \lambda_h f_{\mathrm{haz}}$

5 Closed-loop update

semi-implicit p_{t+1}, q_{t+1}

gate affects material exposure

$g = 0$ recovers geometry-only

6 Tail-sensitive learning

roll out trajectory cost J

optimize $\mathrm{CVaR}_\alpha(J)$

focus updates on worst rollouts